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Adaptive subject-based feature extraction in brain–computer interfaces using wavelet packet best basis decomposition

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Abstract

In this paper we discuss a subject-based feature extraction method using wavelet packet best basis decomposition (WPBBD) in brain–computer interfaces (BCIs). The idea is to employ the wavelet packet best basis algorithm to adapt to each subject separately. Firstly, original electroencephalogram (EEG) signals are decomposed to a given level by wavelet packet transform. Secondly, for each subject, the best basis algorithm is used to find the best-adapted basis for that particular subject. Finally, subband energies contained in the best basis are used as effective features. Adaptive and specific features of a subject are so obtained. Three different motor imagery tasks of six subjects are discriminated using the above features. Experiment results show that the subject-based adaptation method yields significantly higher classification performance than the non-subject-based adaptation and non-adaptive approaches.
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Keywords: Brain–computer interface (BCI); Wavelet packet best basis decomposition (WPBBD); Subject-based feature extraction; Adaptive feature extraction; Electroencephalogram (EEG)

1. Introduction

A brain–computer interface (BCI) is an alternative communication and control channel that does not depend on the brain's normal output pathway of peripheral nerves and muscles [1]. A BCI system is intended to help severely disabled people to communicate with computers or control electronic devices through their thoughts. The most common BCI systems are based on the analysis of spontaneous EEG signals [1]. Over the last years evidence has accumulated to show the possibility to recognize a few mental tasks from spontaneous EEG signals [2–5]. Despite some successful demonstrations of BCIs, there are a number of issues need to be solved before this technology can move outside the lab, especially to fulfill the hopes of disabled people [2]. One of the issues is how to effectively recognize distinguishable EEG signals and so to differentiate human intentions.

The EEG recognition procedure mainly includes the feature extraction and the classification, in which the feature extraction plays an important role for the classification. This paper mainly focuses on the feature extraction. At present, feature extraction methods for spontaneous EEG mainly include such as fast Fourier transform (FFT), autoregressive model (AR) [6–9], wavelet transform and wavelet packet transform (WPT) [10,11]. Due to the non-stationary property of EEG signals [12,13], the WPT can better represent and analyze the signals than FFT or AR as it describes the information in various time windows and frequency bands [14]. As a result of the various time windows, the WPT is capable of capturing non-stationary information such as frequency variation and magnitude undulation but the FFT or AR cannot. The WPT can be viewed as a generalized version of the wavelet transform and it is also demonstrated to outperform other methods for the feature extraction of EEG signals [11].

The existing WPT method in BCIs uses subband energies contained in the last decomposition level as features. It has two disadvantages. One disadvantage is that it is a non-adaptive approach. WPT includes multiple bases and different basis will result in different performance. The best basis can provide "best" classification performance for a

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specified signal. However, the existing WPT method does not perform an adaptive optimization to select the best basis. The other disadvantage is that it is a non-subject-based approach. From the physiological standpoint, no two people are the same and so EEG signals vary between subjects. For instance, it is well known that there exists considerable variability in alpha frequency between subjects [2,15]. We cannot expect all subjects have the same kind of features and these features have the same "best" discrimination power for all of them.

We will discuss an adaptive subject-based method using wavelet packet best basis decomposition (WPBBD). The idea is to adaptively select the best basis fitted for each object. After selecting the best basis, subband energies contained in the best basis are used as features. It should be noted that the number of features extracted from a subject depends on the number of subbands that constitute the selected best basis. The extracted features are fed into a classifier. Three different motor imagery tasks of six subjects are discriminated using the above features. The experiment results show that the subject-based adaptation method yields significantly higher classification performance than the non-subject-based adaptation and non-adaptive approaches.

2. Experimental data

Six healthy subjects (three males, three females; age range, 22–34 years; average age, 27 years) who have no any previous experience with BCIs participated the experiment. They signed informed consent. Subjects seated in a shielded room with dim lighting. Sounds around the surroundings were not controlled painstakingly considering for further application. A 32-channel elastic electrode cap was used to record EEG. The data were recorded at the sampling rate 100 Hz with ESI-128. ESI-128 is a powerful and flexible system for measuring EEG signal up to 128-channel, which is a product of NeuroScan company, US.

Each subject repeated the experiment for two sessions. Each session comprised 5 runs with 30 trials each resulting in a set of 150 trials. The subjects were asked to imagine performing one of three motor imagery tasks (playing basketball using left hand, playing basketball using right hand and braking using right foot) in a self-paced mode during each trial. The number of trials for each task is equal. Each trial lasted 5.75–6.25 s (mean 6 s) and consisted of three phases (shown in Fig. 1): (1) a 0.75–1.25 s (random) resting phase during which the computer screen was black; (2) a 1 s preparation phase during which a "+" fixation was displayed; (3) a 4 s of motor imagery task phase during which the subjects were performing the corresponding motor imagery task according to the direction of the arrow (a left arrow indicating imagining left hand, a right arrow indicating imagining right hand, a down arrow indicating imagining right foot). The arrow was displayed during the first 1 s of the 4 s task phase and

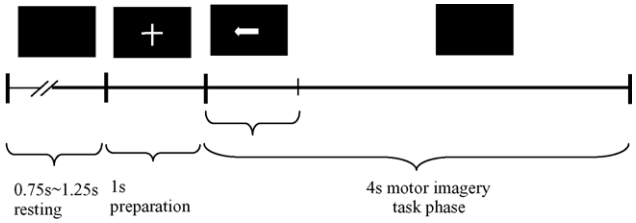


Fig. 1. Three phases of a trial: a resting phase, a preparation phase and a task phase.

the computer screen was black during the other 3 s. The data during the last 4 s of each trial were used to perform off-line analysis.

3. Method

3.1. Wavelet packet transform

The wavelet packet transform (WPT) represents a generalization of the wavelet methods and it has recently been applied to various science and engineering fields with great success [14,16,17]. The WPT is implemented by means of a filter bank and its structure is shown in Fig. 2, where $S(0, 0)$ denotes the original signal space, $S(j, k)$ the decomposed subspace, j the decomposition level and k is the index of the subspace occurring at the jth level. The WPT offers many alternative signal decompositions, such as $S(1,0) \vee S(1, 1)$, $S(2, 0) \vee S(2, 1) \vee S(1, 1)$ and shadowed subspaces. Let $G = \{X_i, i = 1, 2, 3, \ldots\}$ be the set of all valid wavelet packet decompositions and each X_i is called a wavelet packet basis. A valid basis requires that the corresponding subspaces completely cover the entire horizontal "range" without vertical overlap.

3.2. The best basis algorithm

The WPT includes multiple bases and different basis will result in different performance. The best basis is the one that can provide the best performance according to some crite-

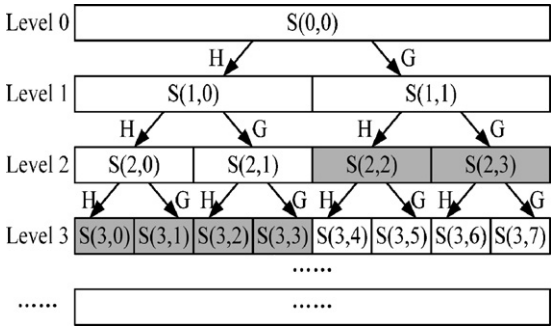


Fig. 2. The structure of WPT (H is a low pass filter and G is a high pass filter).

rion. The criterion is purpose dependent. Entropy is a good measure of information cost and it is suitable for measuring compression effect. However, it may not reflect the classification ability of a basis very well. Our aim is to classify three different motor imagery tasks. Therefore, the separability among classes should be used to measure the efficiency of a basis. We adopts local discriminant basis (LDB) as criterion, which is developed by Saito and Coifman [18]. The goal is to search for an optimal basis X^* with maximum classification distance among wavelet packet library G.

Formally, let us consider a C class classification problem [19]. Let $\chi \subset R^n$ denote a signal space and $\gamma = \{1, 2, \ldots, c, \ldots, C\}$ be a set of the class which the input signals belong. Let $X = \{(x_i, y_i)\}_{i=1}^{N} \subset \chi \times \gamma$ be a training dataset with N pairs of input signals x_i and responses y_i. Let N_c be the number of signals belonging to class c so that we have $N = N_1 + \cdots + N_c$. Also, let us denote a set of class c signals by $\{x_i^{(c)}\}_{i=1}^{N_C}$. In the WPD, we denote $W_{j,k,l}(x_i^{(c)})$ as the decomposition coefficients of signal $x_i^{(c)}$ at subspace $S(j, k)$, where l is the index of the location of decomposition coefficients. Suppose the dimension of signals is $n = 2^{n_0}$, then $l = 0, 1, \ldots, 2^{n_0-j} - 1$. The best basis X^* can be expressed as:

$$X^* = \arg \max_{S(j,k) \in G} H(S(j, k)) \tag{1}$$

where

$$H(S(j, k)) = \sum_{c=1}^{c-1} \sum_{m=c+1}^{c} D(e^{(c)}(j, k), e^{(m)}(j, k)) \tag{2}$$

$$e^{(c)}(j, k) = [e^{(c)}(j, k, 1); e^{(c)}(j, k, 2); \ldots; e^{(c)} \times (j, k, 2^{n_0-j} - 1)] \tag{3}$$

$$e^{(c)}(j, k, l) = \frac{\sum_{i=1}^{N_c} (W_{j,k,l}(x_i^c))^2}{\sum_{i=1}^{N_c} ||x_i^c||^2} \tag{4}$$

$H(S(j, k))$ is called the discriminant power of the subspace $S(j, k)$, $e^{(c)}(j, k)$ is the normalized energy vector of class c, D is Euclidean distance. The best subspace can be determined according to the value of $H(S(j, k))$. As H satisfies the additive property, the best basis X^* can be obtained by using the bottom-up search strategy [18].

3.3. The subject-based feature extraction

The procedure of the subject-based feature extraction using WPBBD can be divided into two parts, in which the first one is for training signals and the second one is for testing signals.

The first one is as follows:

(1) select a wavelet function and specify the decomposition level;
(2) select one subject;
(3) select one of EEG channels from the subject for analysis;
(4) decompose all training signal $(x_i)_{i=1}^{N}$ of the selected channel to the specified level using the selected wavelet and so decomposition coefficients $W_{j,k,l}(x_i)$ can be obtained;
(5) calculate the discriminant power $H(S(j, k))$ for each subspace $S(j, k)$ according to formula (2)–(4);
(6) determine the best basis X^* using the bottom-up search strategy;
(7) calculate the normalized subband (subspace) energy E_j^k of a signal x_i for each subspace $S(j, k)$ that contained in the best basis X^* and select these energies as features, where

$$E_j^k = E(S(j, k)) = \frac{\sum_{l=0}^{2^{n_0-j}-1} [W_{j,k,l}(x_i)]^2}{\sum_{i=1}^{n} ||x_i||^2} \tag{5}$$

As an example, if $X^* = S(2, 0) \vee S(2, 1) \vee S(1, 1)$, we should calculate E_2^0, E_2^1, E_1^1 and use them as features.
(8) repeat steps (3–7) for all EEG channels of the subject;
(9) combine features from all used EEG channels to form the feature vector F. As an example, we assume that two EEG channels (C3 and C4) are used and their best bases are $S(2, 0) \vee S(2, 1) \vee S(1, 1)$ and $S(3, 0) \vee S(3, 1) \vee S(3, 2) \vee S(3, 3) \vee S(2, 2) \vee S(2, 3)$, respectively, then the feature vectors of C3 and C4 channels are $F(C3) = \{E(2, 0), E(2, 1), E(1, 1)\}$ and $F(C4) = \{E(3, 0), E(3, 1), E(3, 2), E(3, 3), E(2, 2), E(2, 3)\}$, respectively. So the final feature vector $F = \{F(C3); F(C4)\}$.
(10) For each subject, repeat steps (2–9).

The algorithm performs an adaptive optimization of the frequency resolution of the overcomplete wavelet packet basis. According to the above procedure, we can find a specific best basis and best feature vector fitted for each object.

The second one is as follows:

(1) select one subject and one EEG channel of the subject;
(2) select a testing signal;
(3) decompose the signal using the same wavelet function and decomposition level as ones used by training signals;
(4) calculate the E_j^k contained in X^* and feature vector F of the testing signal, in which the structures of X^* and F are the same as the ones obtained in training process.

The obtained feature vector F is fed into a classifier. The general flow chart of subject-based feature extraction and classification with the WPBBD is shown in Fig. 3, in which the left of the flow chart is for training signals and the right is for testing signals.

3.4. Parameters selection

3.4.1. Electrode position

The aim of the analysis is to differentiate three motor imagery tasks and so to recognize human's different intentions. Considering the practicality of BCIs system, we use

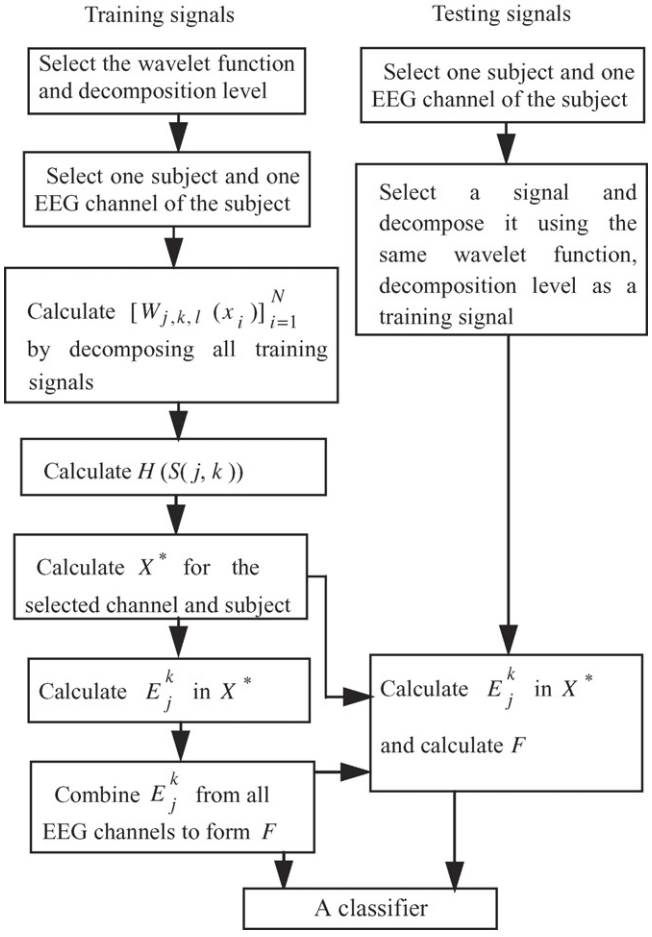


Fig. 3. The general flow chart of the subject-based feature extraction with the WPBBD.

a few electrodes (C3, C4, P3, P4, C_z and P_z electrodes over the sensorimotor cortex, the electrodes position can be seen in ref. [20]). The main reason that these few electrodes are selected is that our cognitive tasks mainly activate local cortical areas below these electrodes and so these few electrodes represent more useful information than other electrodes [2].

3.4.2. Wavelet function and decomposition level

We firstly use the typical "db4" Daubechies wavelet to decompose the training set to five levels. The 100 Hz sample rate and five levels decomposition result in a 1.5625 Hz ($50/2^5 = 1.5625$ Hz) frequency resolution.

3.4.3. Classifier

By virtue of easy training and a solid statistical foundation of the Probabilistic Neural Network (PNN) in Bayesian estimation theory, it has become an effective tool for solving many classification problems [21]. So we use the PNN as our classifier. In the PNN, we take the value of the spread of the radial basis function 1.0.

4. Results and analysis

4.1. Results

In the three-class classification problem, we randomly select one-third of samples from each class for training and the remaining for test. In order to prove the performance of the subject-based adaptation method, we compare it with the non-subject-based adaptation and non-adaptive approaches. These approaches can be described as follows:

(1) *Subject-based adaptation*: The method constructs a wavelet packet best basis fitted for each object. As an example, the best basis and the number of subbands contained in the best basis of subject "1" is shown in Table 1. We can obtain from Table 1 that the dimension of feature vector of subject "1" is 86 (17 + 11 + 14 + 14 + 17 + 13).
(2) *Non-subject-based adaptation*: The method constructs a uniform wavelet packet best basis for all objects. The method also can be called a global adaptation method. We make training samples from all subjects as whole training samples. The best basis is selected using the whole training samples. All subjects use the common best basis and subband energies contained in the best basis are used as features.
(3) *Non-adaptive*: this method is introduced in ref. [11]. Subbands are not selected adaptively and subband energies contained in the last level of wavelet packet decomposition are used as features directly. So, the dimension of the feature vector is 192 (32 (features per channel) × 6 (channels)).

The feature vectors obtained by the above three methods are fed into the PNN classifier, respectively. The classification

Table 1
The best basis of subject "1"

Electrode position	The best basis of the corresponding electrode (channel)	Number of subbands
C3	$S(3,0) \vee (S(5,4)–S(5,9)) \vee S(4,5) \vee (S(5,12)–S(5,17)) \vee S(4,9) \vee S(3,5) \vee S(2,3)$	17
C4	$S(3,0) \vee (S(5,4)–S(5,7)) \vee S(3,2) \vee (S(5,12)–S(5,15)) \vee S(1,1)$	11
P3	$S(3,0) \vee (S(4,2)–S(4,3)) \vee S(3,2) \vee (S(5,12)–S(5,17)) \vee S(4,9) \vee (S(3,5) –S(3,7))$	14
P4	$S(3,0) \vee (S(4,2)–S(4,5)) \vee (S(5,12)–S(5,17)) \vee S(4,9) \vee S(3,5) \vee S(2,3)$	14
C_z	$S(3,0) \vee S(4,2) \vee S(4,3) \vee S(3,2) \vee (S(5,12)–S(5,17)) \vee (S(4,9)–S(4,1,5))$	17
P_z	$(S(3,0)– S(3,2)) \vee S(4,5) \vee (S(5,12)–S(5,17)) \vee S(4,9) \vee S(3,5) \vee S(23)$	13

$S(j,k)$–$S(j,m)$ represents continuous subspaces $S(j,k) \vee S(j,k+1) \vee S(j,k+2) \vee \ldots \vee S(j,m)$, such as $S(5,12)$–$S(5,17)$ means $S(5,12) \vee S(5,13) \vee S(5,14) \vee S(5,15) \vee S(5,16) \vee S(5,17)$.

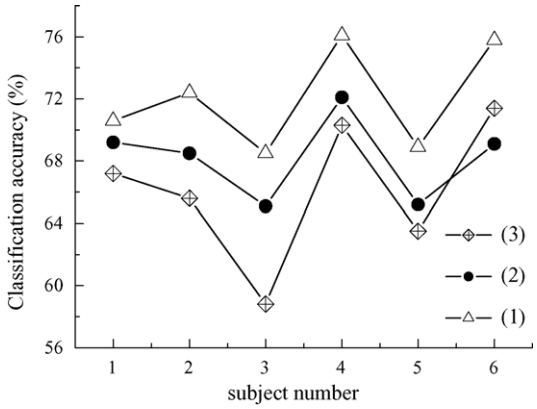


Fig. 4. Classification accuracies of different subjects with different methods: (1) subject-based adaptation; (2) non-subject-based adaptation; (3) non-adaptive.

accuracies are obtained by 10 Cross-Validation. The classification accuracy η is defined as the quotient between the correct samples N_R being classified and all samples N being classified, that is

$$\eta = \frac{N_R}{N} \times 100\% \quad (6)$$

It should be noted that a number of well-known wavelet functions are considered in this study: Haar, Daubechies (db4, db6), Symlet (sym4, sym8). The best result is obtained by 'db4' wavelet function. Fig. 4 plots the classification accuracies of test samples for different subjects with db4 wavelet function. By averaging six subjects, the classification accuracies obtained by different wavelet functions with the methods (1–3) are listed in Table 2.

4.2. Analysis

It can be seen from Fig. 4 and Table 2 that the subject-based adaptation yields the best performance among the three methods. It constructs a wavelet packet best basis fitted for each object and provides the most suitable signal representation for each subject. Therefore it can make each subject obtain the best result. The non-subject-based adaptation constructs a uniform wavelet packet best basis for all objects. So it yields a better performance than the non-adaptive method. However, it cannot ensure all subjects obtain a good performance owing to the difference between subjects. At some time, it is inferior to the non-adaptive method.

Table 2
Average classification accuracy (%) with different wavelet functions and different methods

Method	Wavelet function				
	Haar	db4	db6	sym4	sym8
1	69.86	72.05	70.81	70.51	70.92
2	66.9	68.15	67.96	67.12	66.54
3	63.51	66.13	64.72	64.31	63.83

Method (1): subject-based adaptation; method (2): non-subject-based adaptation; method (3): non-adaptive. db4 and db6 are Daubechies wavelets; sym4 and sym8 are Symlets wavelets.

Considering the limited experimental data, we use all data from two sessions while evaluating the classification performance by 10 Cross-Validation. Samples are thus not independent in the sense that data collected from given experimental sessions are used both for training and testing the classifier. A better approach is to train the classifier on data from one or more sessions and test it on data from different sessions, which would be more realistic and actual for establishing the model of BCIs. We are now trying this approach by analyzing experiments which contain more data and more sessions. In addition, the ultimate goal of BCIs is to realize its actual use. Therefore, an even better approach would be to use the obtained features and classifier on-line in a subsequent closed-loop experiment. Based on the very promising results we obtained here, the on-line performance of the proposed method will be investigated further.

5. Conclusions

(1) The overcomplete structure of WPT provides the flexibility for signal representation to achieve better classification accuracy.
(2) The best basis provides the most suitable frequency subbands for signal representation.
(3) The subject-based adaptation feature extraction with WPBBD construct a wavelet packet best basis fitted for each object and so it can find the suitable and specific features for a subject's EEG signals.

The experiment result also shows that the subject-based adaptation can provide more accurate information and yields better performance than the non- subject-based adaptation and non-adaptive method. It is a promising method and provides a new way for EEG feature extraction in BCIs.

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